

Quick Check

Lesson 3: Why Can a Lone Pawpaw Tree Flower Heavily but Never Set Fruit? Adaptations That Prevent Self-Pollination

9 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.1: Reproduction in Plants | Lesson 3

A pawpaw tree stands alone in a compound in Kisumu. It has hundreds of flowers every season but never sets a single fruit. Which explanation is best supported by what we know about pawpaw?

- A** Pawpaw trees are usually either male or female, so a lone tree may have no partner tree to supply pollen or ovules
- B** Pawpaw flowers always pollinate themselves, so the tree must be sick
- C** Pawpaw trees only produce fruit when grown from cuttings, not from seed
- D** The tree has too many flowers, so none can develop into fruit

In a maize plant, the tassels shed their pollen before the silks on the same plant have emerged and are ready to receive pollen. What is this timing adaptation called?

- A Dioecy
- B Protandry
- C Protogyny
- D Heterostyly

A 'pin' flower has a long style and short stamens. A 'thrum' flower has a short style and long stamens. How does this arrangement (heterostyly) promote cross-pollination?

- A** The stigma of each flower chemically rejects its own pollen
- B** The anthers and stigma are at different heights, so an insect carries pollen from a flower of one type to the stigma of the other type
- C** The pollen is shed before the stigma is ready
- D** The flowers are on separate trees, so pollen must travel between trees

Maize has separate male flowers (tassels) and female flowers (cobs with silks) on the same plant. Does this arrangement completely prevent self-pollination?

- A** Yes, because each maize plant is either male or female
- B** Yes, because male and female flowers can never meet
- C** No, because maize flowers have both male and female parts in the same flower
- D** No, pollen from the tassel can still land on silks of the same plant, so self-pollination is only reduced

A passion fruit flower has mature pollen and a receptive stigma. Pollen from the same flower lands on its stigma, but the pollen does not germinate and is rejected. Which adaptation is this?

A Protandry

B Self-incompatibility

C Dioecy

D Heterostyly

A farmer in Kisii plants cassava cuttings and gets new plants without any flowers or partner plants. Why do cassava cuttings not need a partner?

- A** Cassava cuttings produce pollen that fertilises its own ovules
- B** Cassava trees are male or female, so cuttings take the correct sex
- C** Cassava has self-incompatibility, so it avoids partners
- D** Growing from cuttings is asexual reproduction, so there is no pollination and the new plant is identical to the parent

Which statement best explains why many flowering plants have adaptations that promote cross-pollination?

- A** Flowers that self-pollinate are weak and always die
- B** Self-pollination never produces seeds in any plant
- C** Cross-pollination mixes the genes of two parents, giving varied offspring that may resist disease better
- D** Cross-pollination produces offspring that are exact copies of the parent

After today's work, a class says they can now explain why a lone tree full of flowers bears no fruit. Which new question follows most logically from their evidence?

- A** Why does a lone tree have flowers but no fruit?
- B** Why can a cassava cutting grow into a whole plant without a partner?
- C** What happens inside the flower after compatible pollen lands on a receptive stigma?
- D** Why are maize tassels and silks on the same plant?

A farmer has one female pawpaw tree. It flowers well but gives no fruit. What should the farmer do to get fruit?

- A** Spray the tree with water so the stigmas can reject their own pollen
- B** Take cuttings from the female tree and plant them next to it
- C** Cut off half of the flowers so the rest can pollinate themselves
- D** Plant a male pawpaw tree nearby so pollen can be carried to the female flowers

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).